CAMPUSMONK

TCS Series

ebook

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TCS Series eBook

Q1. Non-Repeating Element

Find the first non-repeating element in a given array arr of N integers.

Note: Array consists of only positive and negative integers and not zero.

Example 1:

Input : arr[] = {-1, 2, -1, 3, 2}

Output : 3

Explanation:

-1 and 2 are repeating whereas 3 is

the only number occuring once.

Hence, the output is 3.

Example 2:

Input : arr[] = {1, 1, 1}

Output : 0

Your Task:

This is a function problem. The input is already taken care of by the driver code. You only need to complete the function firstNonRepeating() that takes an array (arr), sizeOfArray (n), and returns the first non-repeating element. The driver code takes care of the printing.

Expected Time Complexity: O(N).

Expected Auxiliary Space: O(N).

Constraints:

1 <= N <= 107

-1016 <= Ai <= 1016

{Ai !=0 }

Answer

Q2. Airport security officials have confiscated several items of the passenger at the security checkpoint. All the items have been dumped into a huge box(array). Each item possessed a certain amount of risk(0,1,2). Here is the risk severity of the item representing an array[] of N number of integer values. The risk here is to sort the item based on their level of risk values range from 0 to 2.

Example 1:

Input:

7----------Value of N

[1,0,2,0,1,0,2]-> Element of arr[0] to arr[N-1], while input each element is separated by new line

Output:

0 0 0 1 1 2 2 -> Element after sorting based on the risk severity.

Answer

Q3. Array Subset of another array

Given two arrays: a1[0..n-1] of size n and a2[0..m-1] of size m. Task is to check whether a2[] is a subset of a1[] or not. Both the arrays can be sorted or unsorted.

Example 1:

Input:

a1[] = {11, 1, 13, 21, 3, 7}

a2[] = {11, 3, 7, 1}

Output:

Yes

Explanation:

a2[] is a subset of a1[]

Example 2:

Input:

a1[] = {1, 2, 3, 4, 5, 6}

a2[] = {1, 2, 4}

Output:

Yes

Explanation:

a2[] is a subset of a1[]

Example 3:

Input:

a1[] = {10, 5, 2, 23, 19}

a2[] = {19, 5, 3}

Output:

No

Explanation:

a2[] is not a subset of a1[]

Your Task:

You don't need to read input or print anything. Your task is to complete the function isSubset() which takes the array a1[], a2[], its size n and m as inputs and return "Yes" if arr2 is subset of arr1 else return "No" if arr2 is not subset of arr1.

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Expected Time Complexity: O(n)

Expected Auxiliary Space: O(n)

Constraints:

1 <= n,m <= 105

1 <= a1[i], a2[j] <= 106

Answer 

Q4. Sort an array of 0s, 1s and 2s

Given an array of size N containing only 0s, 1s, and 2s; sort the array in ascending order.

Example 1:

Input:

N = 5

arr[]= {0 2 1 2 0}

Output:

0 0 1 2 2

Explanation:

0s 1s and 2s are segregated

into ascending order.

Given an array of size N containing only 0s, 1s, and 2s; sort the array in ascending order.

Example 1:

Input:

N = 5

arr[]= {0 2 1 2 0}

Output:

0 0 1 2 2

Explanation:

0s 1s and 2s are segregated

into ascending order.

Expected Time Complexity: O(N)

Expected Auxiliary Space: O(1)

Constraints:

1 <= N <= 10^6

0 <= A[i] <= 2

Answer 

Q5. Frequencies of Limited Range Array Elements

Given an array A[] of N positive integers which can contain integers from 1 to P where elements can be repeated or can be absent from the array. Your task is to count the frequency of all elements from 1 to N.

Note: The elements greater than N in the array can be ignored for counting and do modify the array in-place.

Example 1:

Input:

N = 5

arr[] = {2, 3, 2, 3, 5}

P = 5

Output:

0 2 2 0 1

Explanation:

Counting frequencies of each array element

We have:

1 occurring 0 times.

2 occurring 2 times.

3 occurring 2 times.

4 occurring 0 times.

5 occurring 1 time.

Example 1:

Input:

N = 5

arr[] = {2, 3, 2, 3, 5}

P = 5

Output:

0 2 2 0 1

Explanation:

Counting frequencies of each array element

We have:

1 occurring 0 times.

2 occurring 2 times.

3 occurring 2 times.

4 occurring 0 times.

5 occurring 1 time.

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Example 2:

Input:

N = 4

arr[] = {3,3,3,3}

P = 3

Output:

0 0 4 0

Explanation:

Counting frequencies of each array element

We have:

1 occurring 0 times.

2 occurring 0 times.

3 occurring 4 times.

4 occurring 0 times.

Your Task:

You don't need to read input or print anything. Complete the function frequencycount() that takes the array and n as input parameters and modify the array itself in place to denote the frequency count of each element from 1 to N. i,e element at index 0 should represent the frequency of 1, and so on.

Can you solve this problem without using extra space (O(1) Space)?

Constraints:

1 ≤ N ≤ 105

1 ≤ P ≤ 4*104

1 <= A[i] <= P

Answer 

Q6. Remove duplicate elements from sorted Array

Given a sorted array A[] of size N, delete all the duplicated elements from A[]. Modify the array such that if there are X distinct elements in it then the first X positions of the array should be filled with them in increasing order and return the number of distinct elements in the array.

Note:

1. Don't use set or HashMap to solve the problem.

2. You must return the number of distinct elements(X) in the array, the generated output will print all the elements of the modified array from index 0 to X-1.

Example 1:

Input:

N = 5

Array = {2, 2, 2, 2, 2}

Output: {2}

Explanation: After removing all the duplicates

only one instance of 2 will remain.

Example 2:

Input:

N = 3

Array = {1, 2, 2}

Output: {1, 2}

Your Task:

You don't need to read input or print anything. Complete the function remove_duplicate() which takes the array A[] and its size N as input parameters and modifies it in place to delete all the duplicates. The function must return an integer X denoting the new modified size of the array.

Expected Time Complexity: O(N)

Expected Auxiliary Space: O(1)

Constraints:

1 ≤ N ≤ 104

1 ≤ A[i] ≤ 106

Answer 

Q7. Find the smallest and second smallest element in an array

Given an array of integers, your task is to find the smallest and second smallest element in the array. If smallest and second smallest do not exist, print -1.

Example 1:

Input :

5

2 4 3 5 6

Output :

2 3

Explanation:

2 and 3 are respectively the smallest

and second smallest elements in the array.

Example 2:

Input :

6

1 2 1 3 6 7

Output :

1 2

Explanation:

1 and 2 are respectively the smallest

and second smallest elements in the array.

Your Task:

You don't need to read input or print anything. Your task is to complete the function minAnd2ndMin() which takes the array A[] and its size N as inputs and returns a vector containing the smallest and second smallest element if possible, else return {-1,-1}.

Expected Time Complexity: O(N)

Expected Auxiliary Space: O(1)

Constraints:

1<=N<=105

1<=A[i]<=105

Answer 

Q8. Binary number to decimal number

Given a Binary Number B, find its decimal equivalent.

Example 1:

Input: B = 10001000

Output: 136

Example 2:

Input: B = 101100

Output: 44

Your Task:

You don't need to read or print anything. Your task is to complete the function binary_to_decimal() which takes the binary number as string input parameter and returns its decimal equivalent.

Expected Time Complexity: O(K * Log(K)) where K is number of bits in binary number.

Expected Space Complexity: O(1)

Constraints:

1 <= number of bits in binary number <= 16

Answer 

Q9. Decimal to binary

Given a decimal number N, compute its binary equivalent.

Example 1:

Input: N = 7

Output: 111

Example 2:

Input: N = 33

Output: 100001

Your Task: You don't need to read input or print anything. Complete the function toBinary() which takes the decimal number N as the input parameter and prints its binary equivalent.

Note: Print the output in a single line, the next line character is printed by the Driver Code.

Answer 

Q10. Searching a number

Given an array Arr of N elements and a integer K. Your task is to return the position of first occurence of K in the given array.

Note: Position of first element is considered as 1.

Example 1:

Input:

N = 5, K = 16

Arr[] = {9, 7, 2, 16, 4}

Output: 4

Explanation: K = 16 is found in the

given array at position 4.

Example 2:

Input:

N = 7, K = 98

Arr[] = {1, 22, 57, 47, 34, 18, 66}

Output: -1

Explanation: K = 98 isn't found in

the given array.

Your Task:

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Complete the function search() which takes an array arr, two integers n and k, as input parameters and returns an integer denoting the answer. Return -1 if the number is not found in array. You don't to print answer or take inputs.

Expected Time Complexity: O(N)

Expected Auxiliary Space: O(1)

Constraints:

1 <= N <= 106

1 <= K <= 106

1 <= Arr[i] <= 106

Answer 

Q11. Binary Search

Given a sorted array of size N and an integer K, find the position(0-based indexing) at which K is present in the array using binary search.

Example 1:

Input:

N = 5

arr[] = {1 2 3 4 5}

K = 4

Output: 3

Explanation: 4 appears at index 3.

Example 2:

Input:

N = 5

arr[] = {11 22 33 44 55}

K = 445

Output: -1

Explanation: 445 is not present.

Your Task:

You dont need to read input or print anything. Complete the function binarysearch() which takes arr[], N and K as input parameters and returns the index of K in the array. If K is not present in the array, return -1.

Expected Time Complexity: O(LogN)

Expected Auxiliary Space: O(LogN) if solving recursively and O(1) otherwise.

Constraints:

1 <= N <= 105

1 <= arr[i] <= 106

1 <= K <= 106

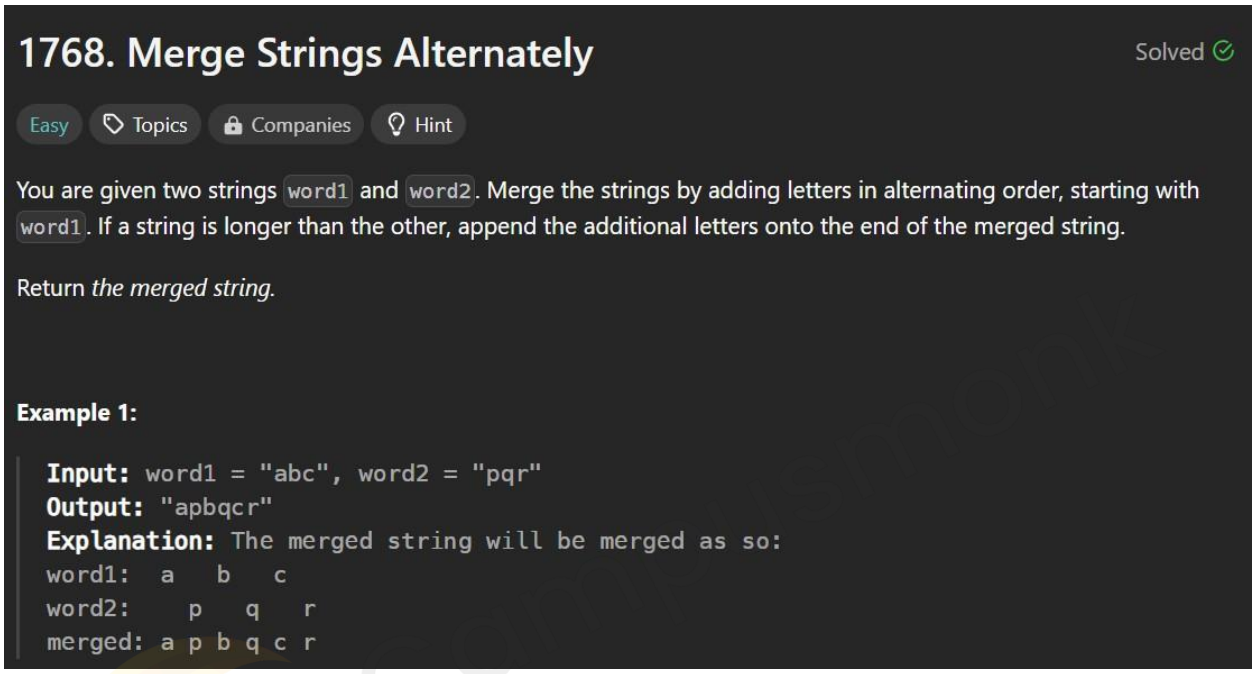
Answer

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LEETCODE 75

ARRAY/STRINGS

DAY-1



1768. Merge Strings Alternately

Solved

Easy | Topics | Companies | Hint

You are given two strings `word1` and `word2`. Merge the strings by adding letters in alternating order, starting with `word1`. If a string is longer than the other, append the additional letters onto the end of the merged string.

Return *the merged string.*

Example 1:

```
Input: word1 = "abc", word2 = "pqr"
Output: "apbqcr"
Explanation: The merged string will be merged as so:
word1:  a   b   c
word2:    p   q   r
merged: a p b q c r
```

https://leetcode.com/problems/merge-strings-alternately/description

```
class Solution {
public:
    vector<bool> kidsWithCandies(vector<int>& candies, int extraCandies) {
        int maxi=INT_MIN;
        for(int i=0;i<candies.size();i++){
            maxi=max(maxi,candies[i]);
        }
        vector<bool>ans(candies.size());
        for(int i=0;i<candies.size();i++){
            if(candies[i]+extraCandies>=maxi){
                ans[i]=true;
            }else{
                ans[i]=false;
            }
        }
```

```
        return ans;
    }
};


class Solution(object):
    def kidsWithCandies(self, candies, extraCandies):
        """
        :type candies: List[int]
        :type extraCandies: int
        :rtype: List[bool]
        """
        max_candi = -float('inf')
        for x in candies:
            max_candi=max(x, max_candi)
        ans=[]
        for x in candies:
            ans.append(True if x+extraCandies >= max_candi else False)
        return ans

class Solution {
    public List<Boolean> kidsWithCandies(int[] candies, int extraCandies) {
        List<Boolean> ans = new ArrayList<Boolean>();
        int maxi = 0;
        for (int i=0;i<candies.length;i++){

            if (candies[i] > maxi){
                maxi = candies[i];
            }
        }

        for (int i=0;i<candies.length;i++){
            if (candies[i]+extraCandies >= maxi){
                ans.add(true);
            } else{
                ans.add(false);
            }
        }
        return ans;
    }

}
```

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1071. Greatest Common Divisor of Strings

Easy | Topics | Companies | Hint

For two strings `s` and `t`, we say "`t` divides `s`" if and only if `s = t + t + t + ... + t + t` (i.e., `t` is concatenated with itself one or more times).

Given two strings `str1` and `str2`, return *the largest string* `x` *such that* `x` *divides both* `str1` *and* `str2`.

Example 1:

> **Input:** `str1 = "ABCABC", str2 = "ABC"`
> **Output:** `"ABC"`

Example 2:

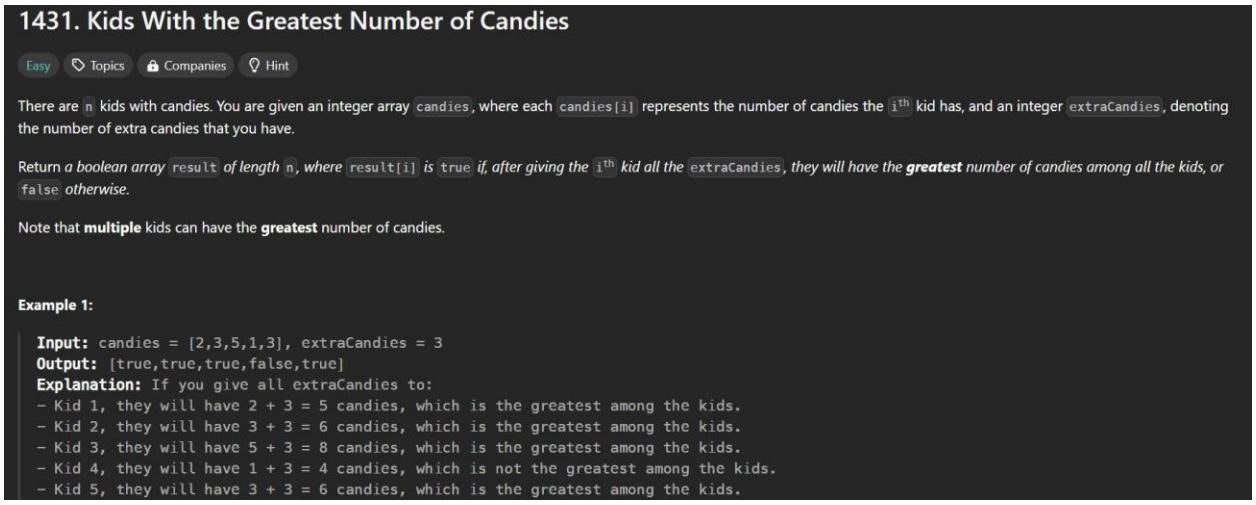
> **Input:** `str1 = "ABABAB", str2 = "ABAB"`
> **Output:** `"AB"`

https://leetcode.com/problems/greatest-common-divisor-of-strings/description

```
class Solution {
public:
    string gcdOfStrings(string str1, string str2) {
        int n = str1.length();
        int m = str2.length();
        int i = 0;
         if(str1 + str2 == str2 + str1){
           int temp  = gcd(n, m);
           return str1.substr(0,temp);
         }
         else{
           return "";
         }
    }
};
```

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1431. Kids With the Greatest Number of Candies

Easy | Topics | Companies | Hint

There are `n` kids with candies. You are given an integer array `candies`, where each `candies[i]` represents the number of candies the `i`th kid has, and an integer `extraCandies`, denoting the number of extra candies that you have.

Return *a boolean array* `result` *of length* `n`*, where* `result[i]` *is* `true` *if, after giving the* `i`th *kid all the* `extraCandies`*, they will have the* ***greatest*** *number of candies among all the kids, or* `false` *otherwise.*

Note that **multiple** kids can have the **greatest** number of candies.

Example 1:

```
Input: candies = [2,3,5,1,3], extraCandies = 3
Output: [true,true,true,false,true]
Explanation: If you give all extraCandies to:
- Kid 1, they will have 2 + 3 = 5 candies, which is the greatest among the kids.
- Kid 2, they will have 3 + 3 = 6 candies, which is the greatest among the kids.
- Kid 3, they will have 5 + 3 = 8 candies, which is the greatest among the kids.
- Kid 4, they will have 1 + 3 = 4 candies, which is not the greatest among the kids.
- Kid 5, they will have 3 + 3 = 6 candies, which is the greatest among the kids.
```

https://leetcode.com/problems/kids-with-the-greatest-number-of-candies/description

```
class Solution {
    public String mergeAlternately(String word1, String word2) {
        int n=word1.length();
        int m=word2.length();

        int i=0;
        int j=0;
        StringBuilder sb=new StringBuilder();
        while(i<n && j<m){
            sb.append(word1.charAt(i));
            sb.append(word2.charAt(j));
            i++;
            j++;
        }

        while(i<n){
            sb.append(word1.charAt(i));
            i++;
        }
        while(j<m){
```

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```
while(i<n){
            sb.append(word1.charAt(i));
            i++;
        }
        while(j<m){
            sb.append(word2.charAt(j));
            j++;
        }
        return sb.toString();
    }

class Solution {
public:
    string mergeAlternately(string word1, string word2) {
        int n = word1.length();
        int m = word2.length();

        int i=0, j=0;


        string result=""; // java


        while (i<n && j<m){
            // first add word1
            result += word1[i]; //java //python
            i++;
            result += word2[j];
            j++;
        }

        while (i<n){
            result += word1[i];
            i++;
        }

        while (j<m){
            result += word2[j];
            j++;
        }

    return result;
    }
};
```

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Home work : Day-1

https://leetcode.com/problems/can-place-flowers/description

https://leetcode.com/problems/reverse-vowels-of-a-string/description

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DAY-2

151. Reverse Words in a String

Solved

Medium | Topics | Companies

Given an input string `s`, reverse the order of the **words**.

A **word** is defined as a sequence of non-space characters. The **words** in `s` will be separated by at least one space.

Return *a string of the words in reverse order concatenated by a single space.*

Note that `s` may contain leading or trailing spaces or multiple spaces between two words. The returned string should only have a single space separating the words. Do not include any extra spaces.

Example 1:

> **Input:** s = "the sky is blue"
> **Output:** "blue is sky the"

Example 2:

> **Input:** s = " hello world "
> **Output:** "world hello"
> **Explanation:** Your reversed string should not contain leading or trailing spaces.

Example 3:

> **Input:** s = "a good   example"
> **Output:** "example good a"
> **Explanation:** You need to reduce multiple spaces between two words to a single space in the reversed

https://leetcode.com/problems/reverse-words-in-a-string/description

```
string reverseWords(string s) {
        stack<string> st;

        string temp="";

        for (int i=0;i<s.size();i++){
            if (i == s.size() or s[i] == ' '){

                if (temp.length())
                    st.push(temp);
                temp="";
            } else{
                temp += s[i]; // blue
            }
        }

        if (temp.length())
```

```
            st.push(temp);

        string ans="";
        while (!st.empty()){
            ans += st.top() + ' ';
            st.pop();
        }

        if (ans[ans.size()-1] == ' ') ans.pop_back();

        return ans;
    }
```

```
class Solution {
    public String reverseWords(String s) {

        String[] str = s.trim().split("\\s+");
        Stack<String> stack = new Stack<>();

        // Push all words onto the stack
        for (String x : str) {
            stack.push(x);
        }

        StringBuilder sb = new StringBuilder();
        // Pop words from the stack to reverse the order
        while (!stack.isEmpty()) {
            sb.append(stack.pop());
            if (!stack.isEmpty()) {
while (!stack.isEmpty()) {
            sb.append(stack.pop());
            if (!stack.isEmpty()) {
                sb.append(" ");
            }
        }

        return sb.toString();
    }
}
```

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238. Product of Array Except Self

Solved

Medium | Topics | Companies | Hint

Given an integer array `nums`, return *an array* `answer` *such that* `answer[i]` *is equal to the product of all the elements of* `nums` *except* `nums[i]`.

The product of any prefix or suffix of `nums` is **guaranteed** to fit in a **32-bit** integer.

You must write an algorithm that runs in `O(n)` time and without using the division operation.

Example 1:

> **Input:** nums = [1,2,3,4]
> **Output:** [24,12,8,6]

Example 2:

> **Input:** nums = [-1,1,0,-3,3]
> **Output:** [0,0,9,0,0]

https://leetcode.com/problems/product-of-array-except-self/description

```
int product = 1;
        int count_zeroes = 0;

        for (int i=0;i<nums.length;i++){
            if (nums[i]!= 0){

                product *= nums[i];
            } else{

                count_zeroes++;
            }
        }

        if (count_zeroes == 0){

            for (int i=0;i<nums.length;i++){
                nums[i] =  product / nums[i];
            }
```

```
        } else if (count_zeroes == 1){

            for (int i=0;i<nums.length;i++){
                if (nums[i] == 0)
                    nums[i] = product;
                else
                    nums[i] = 0;
            }

        } else {

            for (int i=0;i<nums.length;i++){
                nums[i] = 0;
            }
        }
        return nums;
    }
```

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334. Increasing Triplet Subsequence

Medium | Topics | Companies

Given an integer array `nums`, return `true` *if there exists a triple of indices* `(i, j, k)` *such that* `i < j < k` *and* `nums[i] < nums[j] < nums[k]`. If no such indices exists, return `false`.

Example 1:

> **Input:** nums = [1,2,3,4,5]
> **Output:** true
> **Explanation:** Any triplet where i < j < k is valid.

Example 2:

> **Input:** nums = [5,4,3,2,1]
> **Output:** false
> **Explanation:** No triplet exists.

Example 3:

> **Input:** nums = [2,1,5,0,4,6]
> **Output:** true
> **Explanation:** The triplet (3, 4, 5) is valid because nums[3] == 0 < nums[4] == 4 < nums[5] == 6.

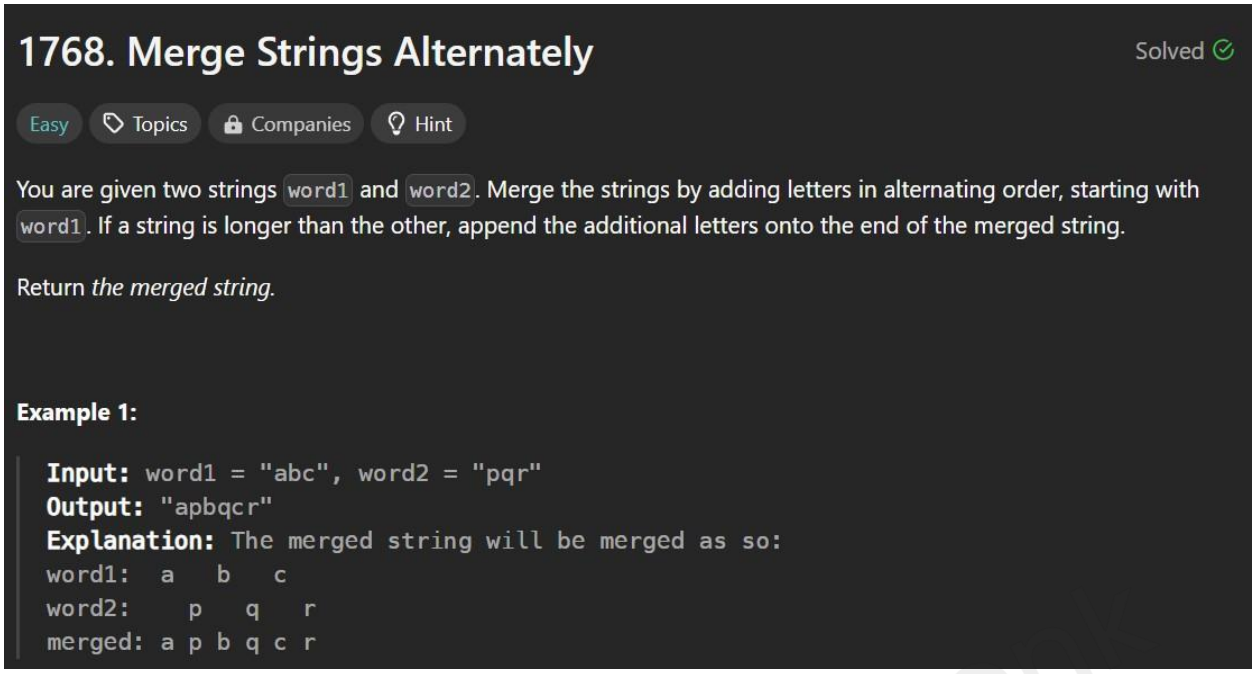
https://leetcode.com/problems/increasing-triplet-subsequence/description/

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HOMEWORK

https://leetcode.com/problems/string-compression/description/

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1768. Merge Strings Alternately

Solved

Easy | Topics | Companies | Hint

You are given two strings `word1` and `word2`. Merge the strings by adding letters in alternating order, starting with `word1`. If a string is longer than the other, append the additional letters onto the end of the merged string.

Return *the merged string.*

Example 1:

> **Input:** word1 = "abc", word2 = "pqr"
> **Output:** "apbqcr"
> **Explanation:** The merged string will be merged as so:
> word1: a b c
> word2: p q r
> merged: a p b q c r

https://leetcode.com/problems/merge-strings-alternately/description

```
class Solution {
public:
    vector<bool> kidsWithCandies(vector<int>& candies, int extraCandies) {
        int maxi=INT_MIN;
        for(int i=0;i<candies.size();i++){
            maxi=max(maxi,candies[i]);
        }
        vector<bool>ans(candies.size());
        for(int i=0;i<candies.size();i++){
            if(candies[i]+extraCandies>=maxi){
                ans[i]=true;
            }else{
                ans[i]=false;
            }
        }

        return ans;
    }
};
```

```
class Solution(object):
    def kidsWithCandies(self, candies, extraCandies):
        """
        :type candies: List[int]
        :type extraCandies: int
        :rtype: List[bool]
```

```
        """
        max_candi = -float('inf')
        for x in candies:
            max_candi=max(x, max_candi)
        ans=[]
        for x in candies:
            ans.append(True if x+extraCandies >= max_candi else False)
        return ans

class Solution {
    public List<Boolean> kidsWithCandies(int[] candies, int extraCandies) {
        List<Boolean> ans = new ArrayList<Boolean>();
        int maxi = 0;
        for (int i=0;i<candies.length;i++){

            if (candies[i] > maxi){
                maxi = candies[i];
            }
        }

        for (int i=0;i<candies.length;i++){
            if (candies[i]+extraCandies >= maxi){
                ans.add(true);
            } else{
                ans.add(false);
            }
        }
        return ans;
    }

}
```

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1071. Greatest Common Divisor of Strings

Easy | Topics | Companies | Hint

For two strings `s` and `t`, we say "`t` divides `s`" if and only if `s = t + t + t + ... + t + t` (i.e., `t` is concatenated with itself one or more times).

Given two strings `str1` and `str2`, return *the largest string* `x` *such that* `x` *divides both* `str1` *and* `str2`.

Example 1:

> **Input:** `str1 = "ABCABC", str2 = "ABC"`
> **Output:** `"ABC"`

Example 2:

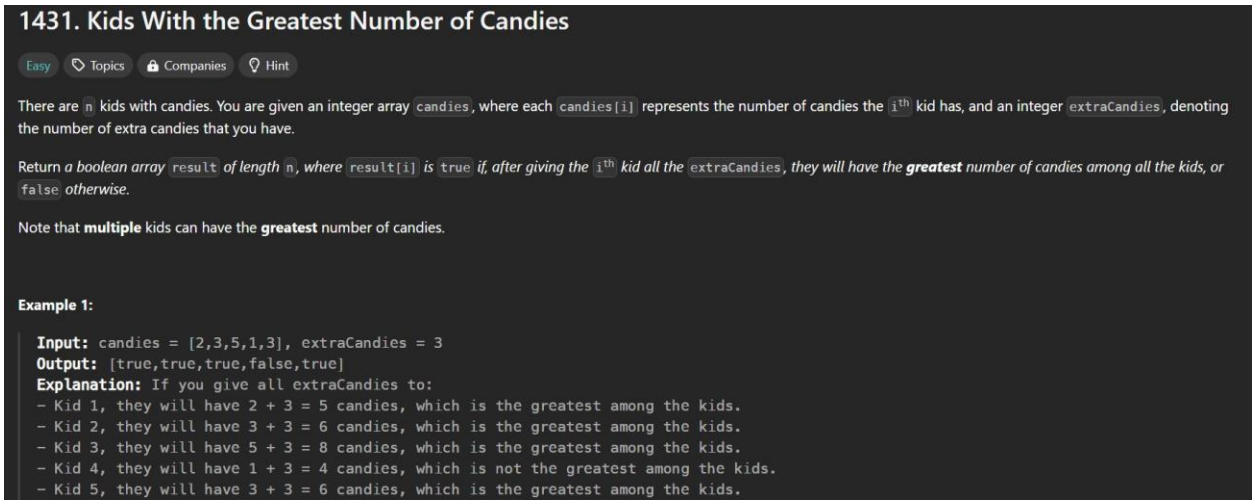
> **Input:** `str1 = "ABABAB", str2 = "ABAB"`
> **Output:** `"AB"`

https://leetcode.com/problems/greatest-common-divisor-of-strings/description

```
class Solution {
public:
    string gcdOfStrings(string str1, string str2) {
        int n = str1.length();
        int m = str2.length();
        int i = 0;
         if(str1 + str2 == str2 + str1){
            int temp  = gcd(n, m);
            return str1.substr(0,temp);
         }
         else{
            return "";
         }
    }
};
```

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1431. Kids With the Greatest Number of Candies

Easy | Topics | Companies | Hint

There are `n` kids with candies. You are given an integer array `candies`, where each `candies[i]` represents the number of candies the `i`th kid has, and an integer `extraCandies`, denoting the number of extra candies that you have.

Return *a boolean array* `result` *of length* `n`*, where* `result[i]` *is* `true` *if, after giving the* `i`th *kid all the* `extraCandies`*, they will have the* ***greatest*** *number of candies among all the kids, or* `false` *otherwise.*

Note that **multiple** kids can have the **greatest** number of candies.

Example 1:

```
Input: candies = [2,3,5,1,3], extraCandies = 3
Output: [true,true,true,false,true]
Explanation: If you give all extraCandies to:
- Kid 1, they will have 2 + 3 = 5 candies, which is the greatest among the kids.
- Kid 2, they will have 3 + 3 = 6 candies, which is the greatest among the kids.
- Kid 3, they will have 5 + 3 = 8 candies, which is the greatest among the kids.
- Kid 4, they will have 1 + 3 = 4 candies, which is not the greatest among the kids.
- Kid 5, they will have 3 + 3 = 6 candies, which is the greatest among the kids.
```

https://leetcode.com/problems/kids-with-the-greatest-number-of-candies/description

```
class Solution {
    public String mergeAlternately(String word1, String word2) {
        int n=word1.length();
        int m=word2.length();

        int i=0;
        int j=0;
        StringBuilder sb=new StringBuilder();
        while(i<n && j<m){
            sb.append(word1.charAt(i));
            sb.append(word2.charAt(j));
            i++;
            j++;
        }

        while(i<n){
            sb.append(word1.charAt(i));
            i++;
        }
        while(j<m){


while(i<n){
             sb.append(word1.charAt(i));
            i++;
        }
        while(j<m){
            sb.append(word2.charAt(j));
            j++;
        }
        return sb.toString();
    }
```

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```
class Solution {
public:
    string mergeAlternately(string word1, string word2) {
        int n = word1.length();
        int m = word2.length();

        int i=0, j=0;


        string result=""; // java


        while (i<n && j<m){
            // first add word1
            result += word1[i]; //java //python
            i++;
            result += word2[j];
            j++;
        }

        while (i<n){
            result += word1[i];
            i++;
        }

        while (j<m){
            result += word2[j];
            j++;
        }

    return result;
    }
};
```

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Home work : Day-1

https://leetcode.com/problems/can-place-flowers/description

```
class Solution {
public:
    bool check_vowel (char x){
        if (x=='a' || x=='e' || x=='i' || x=='O' || x=='U'|| x=='A' || x=='E'
|| x=='I' || x=='o' || x=='u')
            return true;
        return false;
    }
    string reverseVowels(string s) {
        int n = s.size();
        int i = 0;
        int j = n-1;

        while(i<j){
            if (!check_vowel(s[j])){
                j--;
            } else if (!check_vowel(s[i])){
                i++;
            } else{
                swap (s[i], s[j]);
                i++;
                j--;
            }
        }
        return s;
    }
};
```

https://leetcode.com/problems/reverse-vowels-of-a-string/description

345. Reverse Vowels of a String

Easy | Topics | Companies

Given a string `s`, reverse only all the vowels in the string and return it.

The vowels are `'a'`, `'e'`, `'i'`, `'o'`, and `'u'`, and they can appear in both lower and upper cases, more than once.

Example 1:

> **Input:** s = "hello"
> **Output:** "holle"

Example 2:

> **Input:** s = "leetcode"
> **Output:** "leotcede"

Constraints:

- `1 <= s.length <= 3 * 10^5`
- `s` consist of **printable ASCII** characters.

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https://leetcode.com/problems/reverse-vowels-of-a-string/description/

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443. String Compression

Medium | Topics | Companies | Hint

Given an array of characters `chars`, compress it using the following algorithm:

Begin with an empty string `s`. For each group of **consecutive repeating characters** in `chars`:

- If the group's length is `1`, append the character to `s`.
- Otherwise, append the character followed by the group's length.

The compressed string `s` **should not be returned separately**, but instead, be stored **in the input character array** `chars`. Note that group lengths that are `10` or longer will be split into multiple characters in `chars`.

After you are done **modifying the input array,** return *the new length of the array*.

You must write an algorithm that uses only constant extra space.

Example 1:

```
Input: chars = ["a","a","b","b","c","c","c"]
Output: Return 6, and the first 6 characters of the input array should be: ["a","2","b","2","c","3"]
Explanation: The groups are "aa", "bb", and "ccc". This compresses to "a2b2c3".
```

Example 2:

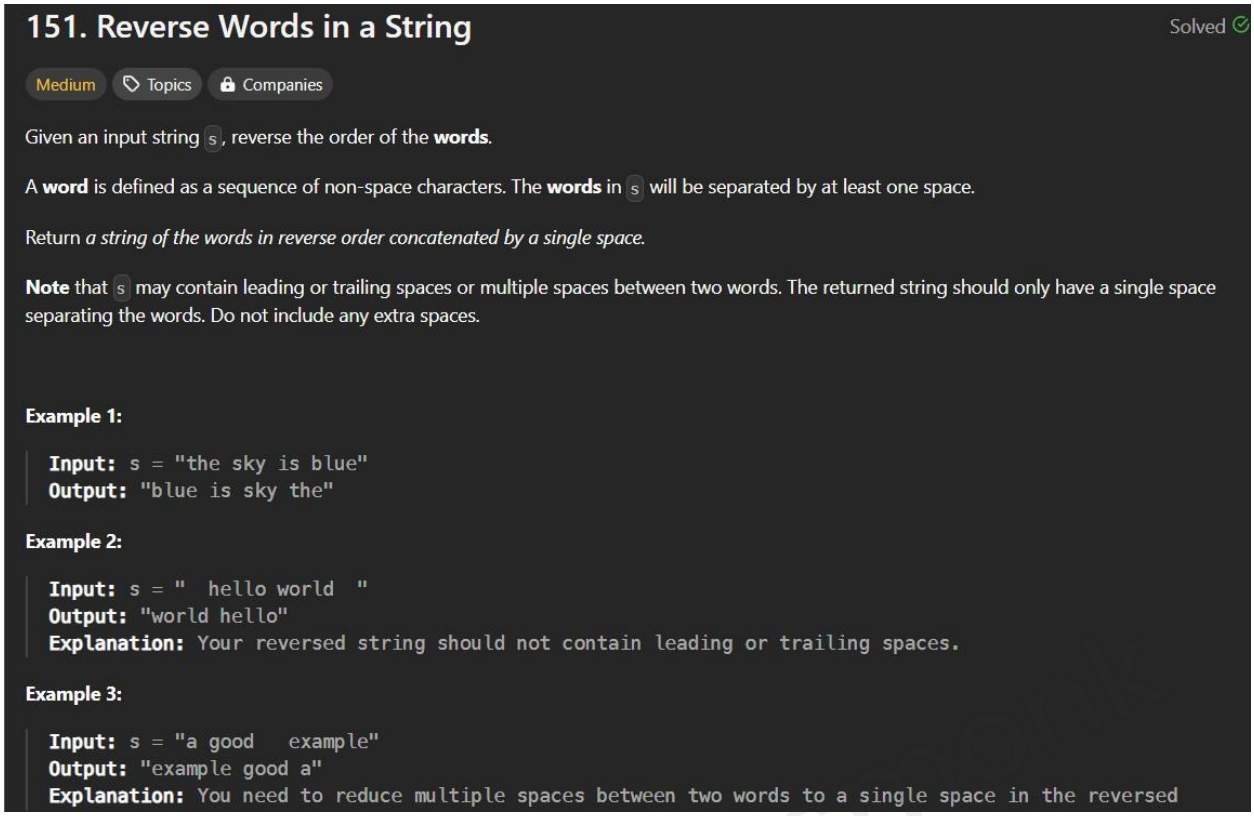
```
Input: chars = ["a"]
Output: Return 1, and the first character of the input array should be: ["a"]
Explanation: The only group is "a", which remains uncompressed since it's a single character.
```

Example 3:

```
Input: chars = ["a","b","b","b","b","b","b","b","b","b","b","b","b"]
Output: Return 4, and the first 4 characters of the input array should be: ["a","b","1","2"].
Explanation: The groups are "a" and "bbbbbbbbbbbb". This compresses to "ab12".
```

https://leetcode.com/problems/string-compression/description/

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151. Reverse Words in a String

Solved

Medium | Topics | Companies

Given an input string `s`, reverse the order of the **words**.

A **word** is defined as a sequence of non-space characters. The **words** in `s` will be separated by at least one space.

Return *a string of the words in reverse order concatenated by a single space.*

Note that `s` may contain leading or trailing spaces or multiple spaces between two words. The returned string should only have a single space separating the words. Do not include any extra spaces.

Example 1:

```
Input: s = "the sky is blue"
Output: "blue is sky the"
```

Example 2:

```
Input: s = "  hello world  "
Output: "world hello"
Explanation: Your reversed string should not contain leading or trailing spaces.
```

Example 3:

```
Input: s = "a good   example"
Output: "example good a"
Explanation: You need to reduce multiple spaces between two words to a single space in the reversed
```

https://leetcode.com/problems/reverse-words-in-a-string/description

```
string reverseWords(string s) {
        stack<string> st;

        string temp="";

        for (int i=0;i<s.size();i++){
            if (i == s.size() or s[i] == ' '){

                if (temp.length())
                    st.push(temp);
                temp="";
            } else{
                temp += s[i]; // blue
            }
        }

        if (temp.length())
            st.push(temp);
```

```
        string ans="";
        while (!st.empty()){
            ans += st.top() + ' ';
            st.pop();
        }

        if (ans[ans.size()-1] == ' ') ans.pop_back();

        return ans;
    }
```

```
class Solution {
    public String reverseWords(String s) {

        String[] str = s.trim().split("\\s+");
        Stack<String> stack = new Stack<>();

        // Push all words onto the stack
        for (String x : str) {
            stack.push(x);
        }

        StringBuilder sb = new StringBuilder();
        // Pop words from the stack to reverse the order
        while (!stack.isEmpty()) {
            sb.append(stack.pop());
            if (!stack.isEmpty()) {
while (!stack.isEmpty()) {
            sb.append(stack.pop());
            if (!stack.isEmpty()) {
                sb.append(" ");
            }
        }

        return sb.toString();
    }
}
```

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605. Can Place Flowers

Easy | Topics | Companies

You have a long flowerbed in which some of the plots are planted, and some are not. However, flowers cannot be planted in **adjacent** plots.

Given an integer array `flowerbed` containing `0`'s and `1`'s, where `0` means empty and `1` means not empty, and an integer `n`, return `true` *if* `n` *new flowers can be planted in the* `flowerbed` *without violating the no-adjacent-flowers rule and* `false` *otherwise.*

Example 1:

> **Input:** flowerbed = [1,0,0,0,1], n = 1
> **Output:** true

Example 2:

> **Input:** flowerbed = [1,0,0,0,1], n = 2
> **Output:** false

Constraints:

- `1 <= flowerbed.length <= 2 * 10^4`
- `flowerbed[i]` is `0` or `1`.
- There are no two adjacent flowers in `flowerbed`.
- `0 <= n <= flowerbed.length`

leetcode.com/problems/can-place-flowers/description/?envType-study-plan-v2&envid=leetcode-75

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392. Is Subsequence

Easy | Topics | Companies

Given two strings `s` and `t`, return `true` *if* `s` *is a* ***subsequence*** *of* `t`, *or* `false` *otherwise.*

A **subsequence** of a string is a new string that is formed from the original string by deleting some (can be none) of the characters without disturbing the relative positions of the remaining characters. (i.e., `"ace"` is a subsequence of `"abcde"` while `"aec"` is not).

Example 1:

> **Input:** s = "abc", t = "ahbgdc"
> **Output:** true

Example 2:

> **Input:** s = "axc", t = "ahbgdc"
> **Output:** false

Constraints:

- `0 <= s.length <= 100`
- `0 <= t.length <= 10^4`
- `s` and `t` consist only of lowercase English letters.

Follow up: Suppose there are lots of incoming `s`, say `s1, s2, ..., sk` where `k >= 10^9`, and you want to check one by one to see if `t` has its subsequence. In this scenario, how would you change your code?

leetcode.com/problems/is-subsequence/?envType=study-plan-v2&envid=leetcode-75

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1732. Find the Highest Altitude

Easy | Topics | Companies | Hint

There is a biker going on a road trip. The road trip consists of `n + 1` points at different altitudes. The biker starts his trip on point `0` with altitude equal `0`.

You are given an integer array `gain` of length `n` where `gain[i]` is the **net gain in altitude** between points `i` and `i + 1` for all (`0 <= i < n`). Return *the* ***highest altitude*** *of a point.*

Example 1:

> **Input:** gain = [-5,1,5,0,-7]
> **Output:** 1
> **Explanation:** The altitudes are [0,-5,-4,1,1,-6]. The highest is 1.

Example 2:

> **Input:** gain = [-4,-3,-2,-1,4,3,2]
> **Output:** 0
> **Explanation:** The altitudes are [0,-4,-7,-9,-10,-6,-3,-1]. The highest is 0.

Constraints:

- `n == gain.length`
- `1 <= n <= 100`
- `-100 <= gain[i] <= 100`

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leetcode.com/problems/find-the-highest-altitude/description/?envType=study-plan-v2&envid=leetcode-75

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283. Move Zeroes

Easy | Topics | Companies | Hint

Given an integer array `nums`, move all `0`'s to the end of it while maintaining the relative order of the non-zero elements.

Note that you must do this in-place without making a copy of the array.

Example 1:

> **Input:** nums = [0,1,0,3,12]
> **Output:** [1,3,12,0,0]

Example 2:

> **Input:** nums = [0]
> **Output:** [0]

Constraints:

- `1 <= nums.length <= 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

Follow up: Could you minimize the total number of operations done?

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leetcode.com/problems/move-zeroes/description/?envType-study-plan-v2&envid=leetcode-75

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11. Container With Most Water

Medium | Topics | Companies | Hint

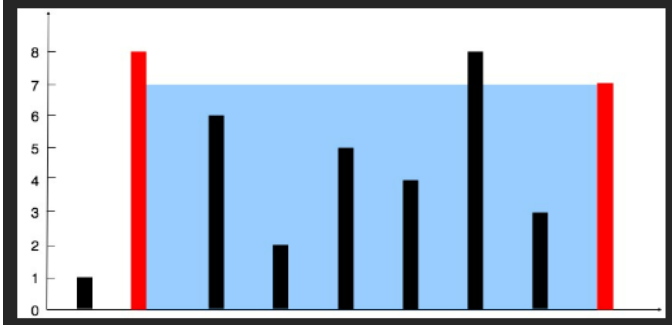
You are given an integer array `height` of length `n`. There are `n` vertical lines drawn such that the two endpoints of the `i^th` line are `(i, 0)` and `(i, height[i])`.

Find two lines that together with the x-axis form a container, such that the container contains the most water.

Return *the maximum amount of water a container can store.*

Notice that you may not slant the container.

Example 1:




```
Input: height = [1,8,6,2,5,4,8,3,7]
Output: 49
Explanation: The above vertical lines are represented by array [1,8,6,2,5,4,8,3,7]. In this case, the max area of water (blue section) the container can contain is 49.
```

Example 2:

```
Input: height = [1,1]
Output: 1
```

Constraints:

- `n == height.length`

leetcode.com/problems/container-with-most-water/description/?envType=study-plan-v2&envid=leetcode-75

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1679. Max Number of K-Sum Pairs

Medium | Topics | Companies | Hint

You are given an integer array `nums` and an integer `k`.

In one operation, you can pick two numbers from the array whose sum equals `k` and remove them from the array.

Return *the maximum number of operations you can perform on the array.*

Example 1:

```
Input: nums = [1,2,3,4], k = 5
Output: 2
Explanation: Starting with nums = [1,2,3,4]:
- Remove numbers 1 and 4, then nums = [2,3]
- Remove numbers 2 and 3, then nums = []
There are no more pairs that sum up to 5, hence a total of 2 operations.
```

Example 2:

```
Input: nums = [3,1,3,4,3], k = 6
Output: 1
Explanation: Starting with nums = [3,1,3,4,3]:
- Remove the first two 3's, then nums = [1,4,3]
There are no more pairs that sum up to 6, hence a total of 1 operation.
```

Constraints:

- `1 <= nums.length <= 10^5`
- `1 <= nums[i] <= 10^9`
- `1 <= k <= 10^9`

leetcode.com/problems/max-number-of-k-sum-pairs/?envType=study-plan-v2&envid=leetcode-75

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643. Maximum Average Subarray I

Easy | Topics | Companies

You are given an integer array `nums` consisting of `n` elements, and an integer `k`.

Find a contiguous subarray whose **length is equal to** `k` that has the maximum average value and return *this value*. Any answer with a calculation error less than 10^{-5} will be accepted.

Example 1:

```
Input: nums = [1,12,-5,-6,50,3], k = 4
Output: 12.75000
Explanation: Maximum average is (12 - 5 - 6 + 50) / 4 = 51 / 4 = 12.75
```

Example 2:

```
Input: nums = [5], k = 1
Output: 5.00000
```

Constraints:

- `n == nums.length`
- `1 <= k <= n <= 10^5`
- `-10^4 <= nums[i] <= 10^4`

leetcode.com/problems/maximum-average-subarray-i/?envType=study-plan-v2&envid=leetcode-75

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1456. Maximum Number of Vowels in a Substring of Given Length

Medium | Topics | Companies | Hint

Given a string `s` and an integer `k`, return *the maximum number of vowel letters in any substring of* `s` *with length* `k`.

Vowel letters in English are `'a'`, `'e'`, `'i'`, `'o'`, and `'u'`.

Example 1:

> **Input:** s = "abciiidef", k = 3
> **Output:** 3
> **Explanation:** The substring "iii" contains 3 vowel letters.

Example 2:

> **Input:** s = "aeiou", k = 2
> **Output:** 2
> **Explanation:** Any substring of length 2 contains 2 vowels.

Example 3:

> **Input:** s = "leetcode", k = 3
> **Output:** 2
> **Explanation:** "lee", "eet" and "ode" contain 2 vowels.

Constraints:

- `1 <= s.length <= 10^5`
- `s` consists of lowercase English letters.
- `1 <= k <= s.length`

leetcode.com/problems/maximum-number-of-vowels-in-a-substring-of-given-length/?envType=study-plan-v2&envid=leetcode-75

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283. Move Zeroes

Easy | Topics | Companies | Hint

Given an integer array `nums`, move all `0`'s to the end of it while maintaining the relative order of the non-zero elements.

Note that you must do this in-place without making a copy of the array.

Example 1:

```
Input: nums = [0,1,0,3,12]
Output: [1,3,12,0,0]
```

Example 2:

```
Input: nums = [0]
Output: [0]
```

Constraints:

- `1 <= nums.length <= 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`



leetcode.com/problems/move-zeroes/description/

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3232. Find if Digit Game Can Be Won

Easy | Topics | Companies | Hint

You are given an array of **positive** integers `nums`.

Alice and Bob are playing a game. In the game, Alice can choose **either** all single-digit numbers or all double-digit numbers from `nums`, and the rest of the numbers are given to Bob. Alice wins if the sum of her numbers is **strictly greater** than the sum of Bob's numbers.

Return `true` if Alice can win this game, otherwise, return `false`.

Example 1:

> **Input:** `nums = [1,2,3,4,10]`
>
> **Output:** `false`
>
> **Explanation:**
>
> Alice cannot win by choosing either single-digit or double-digit numbers.

Example 2:

> **Input:** `nums = [1,2,3,4,5,14]`
>
> **Output:** `true`
>
> **Explanation:**
>
> Alice can win by choosing single-digit numbers which have a sum equal to 15.

Example 3:

> **Input:** `nums = [5,5,5,25]`
>
> **Output:** `true`
>
> **Explanation:**
>
> Alice can win by choosing double-digit numbers which have a sum equal to 25.

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leetcode.com/problems/find-if-digit-game-can-be-won/description/

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3233. Find the Count of Numbers Which Are Not Special

Medium | Topics | Companies | Hint

You are given 2 **positive** integers `l` and `r`. For any number `x`, all positive divisors of `x` *except* `x` are called the **proper divisors** of `x`.

A number is called **special** if it has exactly 2 **proper divisors**. For example:

- The number 4 is *special* because it has proper divisors 1 and 2.
- The number 6 is *not special* because it has proper divisors 1, 2, and 3.

Return the count of numbers in the range `[l, r]` that are **not special**.

Example 1:

> **Input:** `l = 5, r = 7`
>
> **Output:** 3
>
> **Explanation:**
>
> There are no special numbers in the range `[5, 7]`.

Example 2:

> **Input:** `l = 4, r = 16`
>
> **Output:** 11
>
> **Explanation:**
>
> The special numbers in the range `[4, 16]` are 4 and 9.

https://leetcode.com/problems/find-the-count-of-numbers-which-are-not-special/description/

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